ETL/ELT Architecture Design with AWS & ML/LLM Integration

Overview

Modern data pipeline architecture supporting both ETL (Extract-Transform-Load) and ELT (Extract-Load-Transform) patterns, optimized for AWS cloud infrastructure with ML/LLM capabilities.

Architecture Components

1. Data Sources

- **Structured**: RDS/Aurora, DynamoDB
- **Semi-structured**: S3 (JSON, CSV, Parquet)
- **Streaming**: Kinesis Data Streams, MSK (Kafka)
- **APIs**: REST/GraphQL endpoints, SaaS applications
- **Unstructured**: Documents, logs, images (for LLM processing)

2. Ingestion Layer (Extract)

Batch Ingestion

- **AWS Glue**: Serverless ETL jobs for scheduled batch processing
- **Lambda**: Event-driven extraction for smaller datasets
- **Data Migration Service (DMS)**: Database replication and CDC

Real-time Ingestion

- **Kinesis Data Streams**: High-throughput streaming data
- **Kinesis Firehose**: Direct S3/Redshift loading with transformation
- **MSK (Managed Kafka)**: Event streaming for complex workflows

3. Raw Data Storage (Landing Zone)

- **S3 Buckets** (primary landing zone)
  - Immutable raw data storage
  - Partitioned by date/source: `s3://raw-data/{source}/{year}/{month}/{day}/`
  - Lifecycle policies for cost optimization
  - Versioning enabled for data lineage

4. Transformation Layer

ETL Pattern (Transform before Load)

- **AWS Glue ETL Jobs**: PySpark-based transformations
- **Lambda**: Lightweight transformations

- **Step Functions**: Orchestration of complex ETL workflows
- **EMR**: Large-scale data processing with Spark/Hadoop

ELT Pattern (Load then Transform)

- **Redshift**: Data warehouse with SQL-based transformations
- **Athena**: Serverless SQL queries on S3 data lake
- **Glue DataBrew**: Visual data preparation

ML/LLM-Specific Transformations

- **SageMaker Processing Jobs**: Feature engineering pipelines
- **Bedrock**: LLM-based data enrichment
  - Text classification and summarization
  - Entity extraction from unstructured data
  - Sentiment analysis
- **Lambda + Bedrock**: Real-time LLM inference for data augmentation

5. Curated Data Storage

Data Warehouse

- **Redshift**: OLAP queries, BI reporting
- **Redshift Spectrum**: Query data across S3 and Redshift

Data Lake

- **S3 + Glue Catalog**: Centralized metadata repository
- **Lake Formation**: Data lake governance and security
- **Iceberg/Delta Lake**: ACID transactions on S3

Feature Store

- **SageMaker Feature Store**: ML features with online/offline storage

6. ML/LLM Integration Points

Model Training Pipeline

```
Raw Data → Feature Engineering → SageMaker Training → Model Registry → Deployment
```

LLM Use Cases

1. **Document Processing**: Extract insights from PDFs, contracts using Bedrock
2. **Data Quality**: LLM-powered validation and anomaly detection
3. **Metadata Generation**: Auto-generate data descriptions and tags
4. **Synthetic Data**: Generate training data using LLMs

5. **Query Assistance**: Natural language to SQL with Bedrock

ML Feature Pipeline

- **Real-time Features**: Kinesis → Lambda → Feature Store
- **Batch Features**: Glue/EMR → S3 → Feature Store

7. Consumption Layer

Analytics

- **QuickSight**: Dashboards and visualizations
- **Athena**: Ad-hoc SQL queries
- **Redshift**: BI tools integration (Tableau, Power BI)

ML/AI Applications

- **SageMaker Endpoints**: Real-time predictions
- **Bedrock**: LLM-powered applications
- **Lambda**: Serverless ML inference

APIs

- **API Gateway + Lambda**: REST APIs for data access
- **AppSync**: GraphQL APIs

Data Flow Patterns

Pattern 1: Batch ETL with ML

```
Source DB → DMS → S3 Raw → Glue ETL → S3 Curated →
  ↓
  → SageMaker Feature Store → Model Training → Predictions
```

Pattern 2: Streaming ELT with LLM

```
Kinesis Stream → Firehose → S3 Raw →
  ↓
  → Athena Transforms → Curated Layer →
  ↓
  → Lambda + Bedrock → Enriched Data → Applications
```

Pattern 3: Hybrid (ETL + ELT)

```
Multiple Sources → Glue/Lambda → S3 Raw →
  ↓ (ETL)                        ↓ (ELT)
  Redshift ← Glue Transform      Athena Queries
  ↓                              ↓
  BI Tools                       ML Pipelines
```

Key AWS Services Summary

Component	AWS Service	Purpose
Ingestion	Glue, Kinesis, DMS, Lambda	Data extraction
Storage	S3, Redshift	Raw & curated data
Transform	Glue, EMR, Athena, Lambda	Data processing
Orchestration	Step Functions, EventBridge, MWAA	Workflow management
ML/LLM	SageMaker, Bedrock	Model training & inference
Governance	Lake Formation, Glue Catalog	Metadata & security
Monitoring	CloudWatch, X-Ray	Observability

ML/LLM-Specific Considerations

Feature Engineering

- Use Glue/EMR for large-scale feature computation
- SageMaker Processing for complex transformations
- Feature Store for serving and versioning

LLM Data Preparation

- **Prompt Engineering**: Store prompt templates in S3
- **Context Windows**: Chunk large documents (Lambda/Glue)
- **RAG Architecture**: Vector embeddings in OpenSearch/RDS pgvector
- **Fine-tuning Data**: Curated datasets in S3 for model customization

Model Serving

- **Batch Predictions**: Glue/EMR → SageMaker Batch Transform
- **Real-time**: API Gateway → Lambda → SageMaker/Bedrock
- **Streaming**: Kinesis → Lambda → Feature Store → Inference

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Data Governance

Security

- **IAM Roles**: Least privilege access
- **KMS Encryption**: At-rest and in-transit
- **VPC Endpoints**: Private connectivity
- **Secrets Manager**: Credentials management

Quality

- **Glue Data Quality**: Automated validation rules
- **Great Expectations**: Custom quality checks
- **LLM-based Validation**: Bedrock for semantic checks

Lineage

- **Glue Catalog**: Table-level lineage
- **Tags**: Business context and ownership
- **CloudTrail**: Audit logs

Orchestration Examples

ETL Workflow (Step Functions)

```
{
  "Extract": "DMS/Glue Crawler",
  "Transform": "Glue ETL Job",
  "Validate": "Lambda + Data Quality",
  "Load": "Copy to Redshift",
  "Notify": "SNS Alert"
}
```

ML Pipeline (SageMaker Pipelines)

```
{
  "DataPrep": "Glue/Processing Job",
  "FeatureEngineering": "SageMaker Processing",
  "Training": "SageMaker Training",
  "Evaluation": "Lambda + Bedrock (model assessment)",
  "Deployment": "SageMaker Endpoint"
}
```

Cost Optimization

1. **S3 Intelligent Tiering**: Auto-move to cheaper storage classes
2. **Redshift RA3**: Separate compute and storage
3. **Glue Auto Scaling**: Dynamic DPU allocation
4. **Spot Instances**: For EMR clusters

5. **Bedrock On-Demand**: Pay per token vs provisioned throughput
6. **Athena**: Query only required partitions

Monitoring & Observability

- **CloudWatch Dashboards**: Pipeline health metrics
- **CloudWatch Alarms**: Failed jobs, SLA breaches
- **X-Ray**: End-to-end tracing for Lambda/API calls
- **Glue Job Metrics**: DPU usage, record counts
- **SageMaker Model Monitor**: Drift detection
- **SNS/Slack**: Alerting

Example Use Cases

1. Customer 360 (ETL)

```
CRM/ERP/Web → Glue → S3 → Redshift → QuickSight
                ↓
              Bedrock (entity resolution, deduplication)
```

2. Real-time Fraud Detection (Streaming + ML)

```
Transactions → Kinesis → Lambda → Feature Store → SageMaker Endpoint → Alert
```

3. Document Intelligence (LLM)

```
PDFs → S3 → Textract → Lambda + Bedrock → Extract entities → Redshift → Search
```

4. Predictive Analytics (Hybrid)

```
Historical Data → S3 → Athena (ELT) → Feature Store →
                                          ↓
                                  SageMaker Training → Inference
```

Technology Stack

- **Languages**: Python (PySpark, Boto3), SQL
- **IaC**: Terraform or CloudFormation
- **Orchestration**: Step Functions, MWAA (Airflow), EventBridge

- **Version Control**: Git, CodeCommit
- **CI/CD**: CodePipeline, GitHub Actions

Next Steps

1. Define specific use case requirements
2. Design detailed data models (schemas)
3. Implement proof of concept for one pipeline
4. Set up monitoring and alerting
5. Document data dictionary and lineage
6. Establish governance policies
7. Train team on AWS services and ML/LLM tools

This design provides a flexible foundation that can scale from simple batch ETL to complex real-time ML/LLM workflows on AWS.